

{'analysis_time': 0.0,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 12.520782550219723,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 12.520782550219723,
'retro_star_value': 12.520782550219723}

Reaction
{'analysis_time': 0.09129,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 12.520782550219723,
'retro_star_reaction_number': 12.520782550219723,
'retro_star_rxn_cost': 2.7168498748467584,
'retro_star_value': 12.520782550219723}

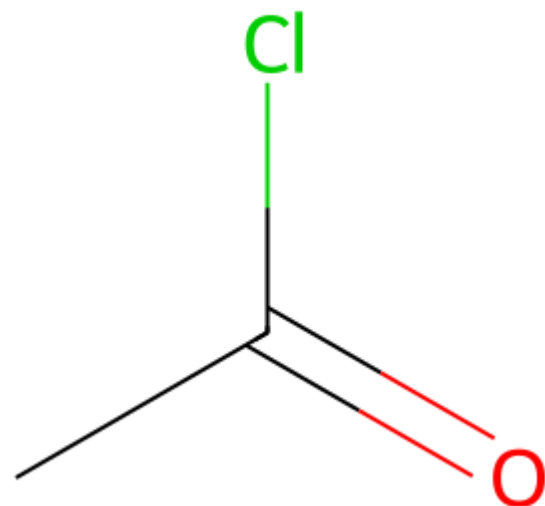
HCl

{'analysis_time': 0.09129,
'first_solution_time': 0.271394,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.5304296406560596,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.5304296406560596,
'retro_star_value': 12.520782550219723}

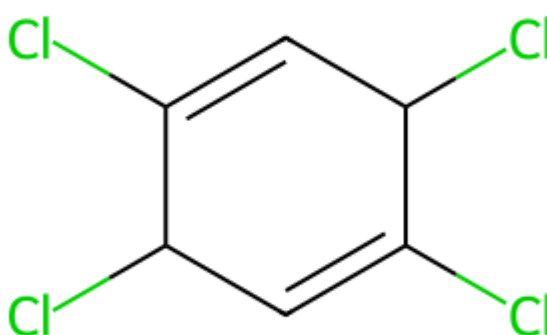
{'analysis_time': 0.09129,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 8.273503034716905,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 8.273503034716905,
'retro_star_value': 12.520782550219723}

Reaction
{'analysis_time': 0.271394,
'first_solution_time': 0.271394,
'num_calls_rxn_model': 2,
'num_calls_value_function': 10,
'retro_star_min_cost': 1.5304296406560596,
'retro_star_reaction_number': 1.5304296406560596,
'retro_star_rxn_cost': 1.5304296406560596,
'retro_star_value': 12.520782550219723}

Reaction
{'analysis_time': 0.357056,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 3,
'num_calls_value_function': 25,
'retro_star_min_cost': 8.957211120448877,
'retro_star_reaction_number': 8.957211120448877,
'retro_star_rxn_cost': 1.4880498912157807,
'retro_star_value': 13.204490635951695}



{'analysis_time': 0.271394,
'first_solution_time': 0.271394,
'num_calls_rxn_model': 2,
'num_calls_value_function': 10,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 12.520782550219723}



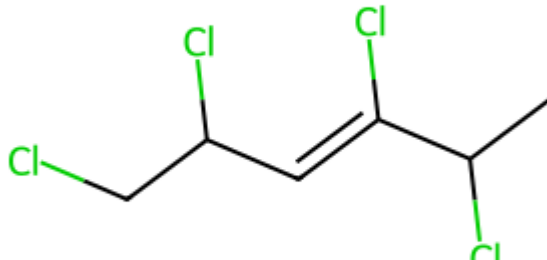
{'analysis_time': 0.357056,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 3,
'num_calls_value_function': 25,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 5.938731588577037,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 5.938731588577037,
'retro_star_value': 13.204490635951695}

HCl

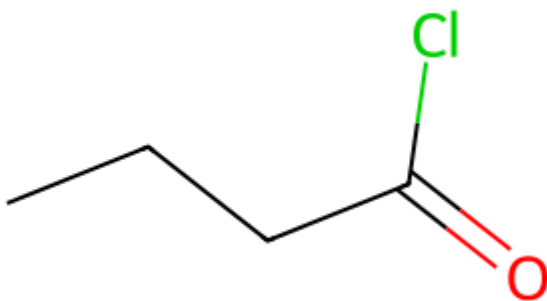
{'analysis_time': 0.357056,
'first_solution_time': 0.360138,
'num_calls_rxn_model': 3,
'num_calls_value_function': 25,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.5304296406560596,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.5304296406560596,
'retro_star_value': 13.204490635951695}

Reaction
{'analysis_time': 0.634956,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 6,
'num_calls_value_function': 230,
'retro_star_min_cost': 5.938731588577037,
'retro_star_reaction_number': 5.938731588577037,
'retro_star_rxn_cost': 1.164047429227592,
'retro_star_value': 13.204490635951695}

Reaction
{'analysis_time': 0.360138,
'first_solution_time': 0.360138,
'num_calls_rxn_model': 3,
'num_calls_value_function': 94,
'retro_star_min_cost': 6.848789537414928,
'retro_star_reaction_number': 6.848789537414928,
'retro_star_rxn_cost': 6.848789537414928,
'retro_star_value': 18.522850532710564}

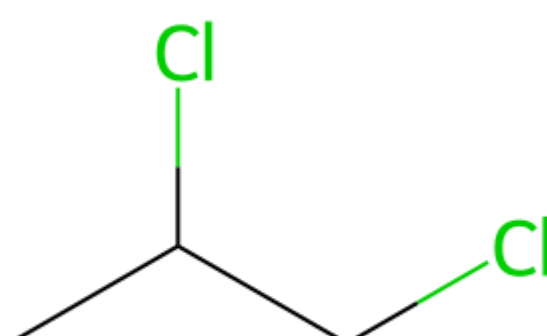


{'analysis_time': 0.634956,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 6,
'num_calls_value_function': 230,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.774684159254278,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.774684159254278,
'retro_star_value': 13.204490635951695}



{'analysis_time': 0.360138,
'first_solution_time': 0.360138,
'num_calls_rxn_model': 3,
'num_calls_value_function': 94,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 18.522850532710564}

Reaction
{'analysis_time': 1.15047,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 12,
'num_calls_value_function': 491,
'retro_star_min_cost': 5.967045841131245,
'retro_star_reaction_number': 5.967045841131245,
'retro_star_rxn_cost': 5.967045841131245,
'retro_star_value': 14.396852317828662}



{'analysis_time': 1.15047,
'first_solution_time': 1.15047,
'num_calls_rxn_model': 12,
'num_calls_value_function': 491,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 14.396852317828662}